POSTER: Learning Heuristics for Multiple Gapped Longest Common Subsequence Problem

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the quality of solutions

Abstract

The Variable Gapped Longest Common Subsequence Problem (VGLCSP) is a challenging combinatorial optimization problem arising in sequence analysis with applications in computational biology and time series analysis, where the search space grows rapidly with the number and with gap constraints. Beam Search–based heuristics can provide good approximate solutions, but their effectiveness strongly depends on the quality of the heuristic used to evaluate partial subsequences.

This work proposes a learning-based framework to automatically construct a heuristic for guiding Beam Search (BS) in the VGLCSP. The approach integrates a feed-forward neural network (FFNN) within the Iterative Multi-Source BS (IMSBS) algorithm, where the network estimates the quality of candidate extensions during the search. Instead of gradient-based training, the network parameters are optimized using a Genetic Algorithm (GA), allowing the heuristic to be learned directly from solution quality obtained during the search process. Compact feature representations are designed to encode the state of partial solutions, enabling the learned heuristic to generalize across instances with varying sequence lengths, numbers of sequences, and alphabet sizes. During training, candidate neural networks are evaluated by embedding them within the BS procedure and measuring solution quality on a set of training instances, while separate validation instances are used to monitor generalization.

Preliminary experiments indicate that the learned heuristic can improve the performance of the baseline IMSBS algorithm on a range of VGLCSP instances. These results demonstrate the potential of combining evolutionary optimization with neural network models to automatically design effective heuristics for complex combinatorial search problems.

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CCS Concepts

- Computing methodologies → Artificial intelligence, Machine learning;
- Theory of computation → Design and analysis of algorithms;

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http://dx.doi.org/10.1145/

Keywords

Beam Search; Genetic Algorithms; Neuro-evolution; Longest Common Subsequence; Bioinformatics; Combinatorial Optimization

ACM Reference Format

Marko Djukanović. 2026. Learning Heuristics for the Multiple Gapped Longest Common Subsequence Problem: Artifact Appendix. University of Nova Gorica.

1. Introduction

The Longest Common Subsequence (LCS) problem and its variants are fundamental problems in sequence analysis and bioinformatics. One particularly challenging variant is the *Variable Gapped Longest Common Subsequence Problem* (VGLCSP) [1], where gap constraints restrict the allowable distances between consecutive matches of solutions in each input sequence. These constraints significantly increase the complexity of the search space and make exact algorithms impractical for medium and large instances.

Heuristic search methods such as Beam Search (BS) have been proposed to handle this complexity by exploring only the most promising candidate solutions. However, the effectiveness of BS strongly depends on the quality of the heuristic used to rank partial solutions. Traditional heuristics are typically handcrafted and remain static across instances, limiting their ability to adapt to diverse problem structures [3].

This work introduces a learning-based framework for automatically constructing such heuristics. The key idea is to integrate machine learning with combinatorial search so that the heuristic function is learned directly from data and optimized according to the final solution quality achieved by the search algorithm.

2. Proposed Approach

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The proposed framework combines three main components:

1. **Iterative Multi-Source Beam Search (IMSBS)** [3] for constructing ~~candidate subsequences~~.
2. **Feed-Forward Neural Network (FFNN)** [4] acting as a heuristic evaluation function for partial solutions.
3. **Genetic Algorithm (GA)** [2] for optimizing the neural network parameters.

In the learning process, each GA individual represents a flattened vector of neural network weights. For each individual, the weights are loaded into the FFNN, which evaluates partial ~~states~~ solutions generated during a BS execution. The IMSBS algorithm then constructs subsequences guided by the neural heuristic. The fitness

Complete solutions

before you call them "parameters"

solutions solution and LIMSBS

of each individual corresponds to the average solution quality obtained on a training set of instances. Validation instances are evaluated separately to monitor generalization performance.

A key aspect of the approach is the design of compact feature representations for partial search states. Each node in the search tree is described by a feature vector containing (i) normalized positions of current matches in each sequence: statistical descriptors (minimum, maximum, mean, and standard deviation) of these positions; (ii) the current length of the partial subsequence; (iii) statistical summaries of remaining gap constraints; (iv) global instance features such as alphabet size $|\Sigma|$. Some other global features have been taken into account such as number of sequences (m) or lenght of input sequences (n), however without improving the results.

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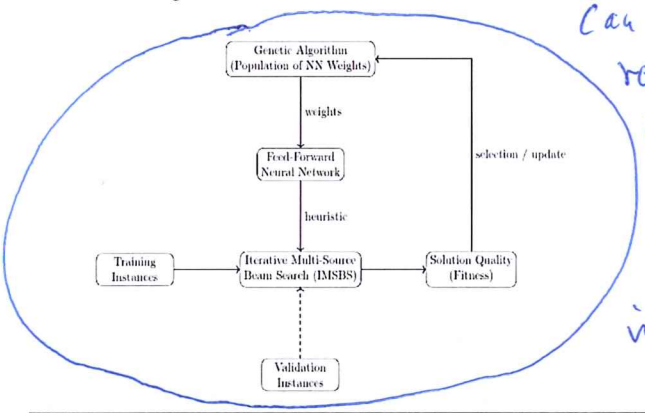
These features provide a low-dimensional yet informative representation of the search state and allow the neural network to generalize across instances of different sizes. The learning process is illustrated in Figure 1.



Can this be regenerated in order to be better visible in such a small space?

Figure 1. Graphical representation of learning a NN-guidance.

3. Experimental Results

The experiments evaluate the performance of the learning-based Beam Search (LIMSBS) against the baseline IMSBS algorithm proposed in [3] on the set of random instances generated in the latter paper. For each combination number of sequences $m \in \{2, 3, 5, 10\}$, sequence length $n \in \{50, 100, 200, 500\}$, and alphabet size $|\Sigma| \in \{2, 4\}$, 10 random instances generated. All sequences are randomly generated. In overall, 320 problem instances are provided. Both algorithms are tuned by grid search. The best-performing neural network configuration uses three hidden layers with 10 neurons each and the ReLU activation function, among the three tested (`ReLU`, `sigmoid`, and `tanh`). Training process to learn neural heuristic using BRKGA is performed with two hours time limit on the High performance computing infrastructure VEGA, placed at IZUM in Maribor, Slovenia, equipped with AMD EPYC 7H12 CPUs operating at 2.35 GHz. Other crucial parameters in both IMSBS and LMSBS are set to beam width $\beta = 1000$ and $imsbs_iters = 1000$.

are of the FFNN training is performed with

Despite this relatively modest training budget, the learned heuristic demonstrates competitive performance and improves average solution quality in several instance categories, particularly for larger numbers of sequences, see numerical results in Table 1. These results indicate learning-based heuristics can effectively guide BS and adapt to structural characteristics of the sequence problems, in this case the VGLCS, especially suited for $m \geq 5$.

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4. Conclusions

This work provides the rough implementation details and experimental framework for learning heuristics in combinatorial sequence optimization using neuro-evolutionary methods. By integrating Beam Search with neural networks optimized via genetic algorithms, the approach enables adaptive heuristic construction in non-differentiable settings.

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The reported results provide a promising basis for further experimentation on learning-guided combinatorial search methods. Beyond the VGLCSP, the proposed framework can be extended to other sequence alignment and combinatorial optimization problems where heuristic design is difficult.

Acknowledgments. This publication is co-funded by the European Union's Horizon Europe research and innovation program under the Marie Sklodowska-Curie COFUND Postdoctoral Programme (grant agreement No. 101081355 – SMASH), and by the Republic of Slovenia and the European Union through the European Regional Development Fund. Views and opinions expressed are those of the authors only and do not necessarily reflect those of the European Union or the European Research Executive Agency (REA). Neither the European Union nor the REA can be held responsible for them.

Table 1. Comparison of objective values between Learning-based BS and IMSBS. Best results per instance category are in bold.

m	n	$\|\Sigma\|$	IMSBS	LIMSBS
2	100	2	**72.9**	72.5
2	100	4	**61.7**	60.3
2	200	2	**139.7**	139.3
2	200	4	**125.2**	120.9
2	500	2	281.9	**295.4**
2	500	4	**302.9**	285.4
2	50	2	**37.7**	**37.7**
2	50	4	**30.1**	29.7
3	100	2	**58.5**	**58.5**
3	100	4	**48.5**	48.0
3	200	2	**106.3**	103.4
3	200	4	**97.4**	95.6
3	500	2	133.5	**143.1**
3	500	4	**230.3**	218.8
3	50	2	**31.2**	**31.2**
3	50	4	**22.9**	**22.9**
5	100	2	**31.5**	31.0
5	100	4	25.9	**26.9**
5	200	2	35.2	**41.6**
5	200	4	34.0	**37.2**
5	500	2	31.0	**39.3**
5	500	4	36.3	**46.2**
5	50	2	**20.5**	**20.5**
5	50	4	**15.3**	**15.3**
10	100	2	10.5	**12.0**
10	100	4	7.6	**8.9**
10	200	2	11.8	**12.2**
10	200	4	7.6	**8.2**
10	500	2	12.5	**12.9**
10	500	4	7.5	**8.3**
10	50	2	**11.4**	11.2
10	50	4	**7.1**	**7.1**
Avg.			65.20	**65.67**

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LIMSBS